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Fastening arrangement and seat

Abstract

A fastening arrangement of a seat on a rail arrangement may have a modular construction and may have one lock module for releasably arresting the fastening arrangement on the rail arrangement. The fastening arrangement may also have one lock unlocking module for unlocking the lock module. The fastening arrangement may also have one rail unlocking module.

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Background/Summary

FIELD

(1) The invention relates to a fastening arrangement of a seat on a rail arrangement and to a seat with such a fastening arrangement.

BACKGROUND

(2) Fastening arrangements for a seat are known in the prior art. For example, the seat can be fastened on a rail arrangement releasably by hooks or screws or nonreleasably by a welded

connection.

SUMMARY

(3) It is the object of the present invention to specify a simple fastening arrangement of a seat on a rail arrangement and a seat with such a simple fastening arrangement.

(4) The object is achieved according to the invention by a fastening arrangement of a seat on a rail arrangement, wherein the fastening arrangement is of modular construction and comprises at least one lock module for releasably arresting the fastening arrangement on the rail arrangement, one lock unlocking module and one rail unlocking module.

(5) Using such a modular seat fastening arrangement, a seat can be simply and rapidly mounted on and removed from a seat longitudinal adjustment device. The individual modules, such as the lock module, the lock unlocking module and the rail unlocking module, can be preassemblable here. Such a modular fastening arrangement has few parts and can be mounted particularly simply and rapidly and also has a low weight.

(6) Furthermore, the lock unlocking module and the rail unlocking module are operable in a simple manner independently of one another or synchronously, i.e. can be brought from an arresting position of a seat lock and/or a rail locking means into a released position of the seat lock or of the rail locking means, or vice versa.

(7) The lock module can be of multi-part design and comprises, for example, two or four lock units, e.g. one pair of lock units or two pairs of lock units. In one possible embodiment, the lock units are synchronously operable, in particular unlockable, by the lock unlocking module.

(8) The lock units are preferably arranged on a common carrier. In particular, the lock units can be preassembled on the common carrier.

(9) Furthermore, the fastening arrangement can comprise an actuating module for actuating the lock unlocking module and/or the rail unlocking module. The actuating module can comprise one or more actuating units for dependently or independently actuating the lock unlocking module and/or the rail unlocking module. The actuating units can be formed separately. However, the actuating units can also be coupled in terms of movement. The actuating units can be designed as a lever arrangement, in particular as a lever arm arrangement, and/or as a tension arrangement, in particular tension strap arrangement or tension belt arrangement.

(10) In one possible embodiment, the actuating module is coupled to the lock module via the lock unlocking module.

(11) Alternatively or additionally, the actuating module can be coupled to a locking element for the rail arrangement via the lock unlocking module.

(12) With regard to the seat according to the invention, the object is achieved by a rail arrangement and the previously described fastening arrangement, which is designed for releasably fastening the seat on the rail arrangement.

(13) In one possible embodiment, the fastening arrangement is designed as a preassemblable docking mechanism. The docking mechanism here comprises the lock module and also the lock unlocking module and the rail unlocking module. Preferably, an unlocking element and/or an unlocking mechanism for the rail arrangement can additionally be part of the preassemblable docking mechanism. Such a modular preassemblable docking mechanism for releasably fastening the seat on the rail arrangement permits simple and rapid installation or removal of the seat on or from the rail arrangement.

(14) In one possible embodiment, the rail arrangement comprises two pairs of rails each having an upper rail and a lower rail. The fastening arrangement is releasably arrestable at one rail end of the upper rail by a respective lock unit. Such an in particular end-face releasable arresting of the fastening arrangement on the rail ends of the upper rail is simply constructed and permits simple and reliable installation of the fastening arrangement on the rail arrangement.

Description

DESCRIPTION OF THE FIGURES

- (1) Exemplary embodiments of the invention will be discussed in more detail with reference to drawings, in which:
- (2) FIG. 1 shows a perspective illustration of a seat with a rail arrangement,
- (3) FIG. 2 shows a schematic illustration of an embodiment of a modular fastening arrangement of a seat on a rail arrangement,
- (4) FIG. 3 schematically shows, in an exploded illustration, a first embodiment of a modular fastening arrangement,
- (5) FIG. 4 schematically shows, in an exploded illustration, a second embodiment of a modular fastening arrangement,
- (6) FIG. 5 schematically shows, in a perspective view, a second embodiment of a modular fastening arrangement,
- (7) FIG. 6 schematically shows, in an exploded illustration, a third embodiment of a modular fastening arrangement,
- (8) FIG. 7 schematically shows, in a perspective view, a third embodiment of a modular fastening arrangement,
- (9) FIG. 8 schematically shows, in an exploded illustration, an exemplary embodiment of a lock unit,
- (10) FIG. 9 schematically shows, in the assembled state, an exemplary embodiment of a lock unit,
- (11) FIG. 10 shows a schematic illustration of a releasable seat on a rail arrangement,
- (12) FIG. 11 shows a schematic illustration of the releasable seat pre-positioned on the rail arrangement, and
- (13) FIG. 12 shows a schematic perspective view of one embodiment of a fastening arrangement in the unlocked or open state,
- (14) FIG. 13 shows a schematic perspective view of one embodiment of an actuating module,
- (15) FIG. 14A shows a schematic side view of one embodiment of a carrier plate of a lock unit,
- (16) FIG. 14B shows a schematic side view of one embodiment of an unlocking mechanism on the carrier plate of a lock unit,
- (17) FIG. 15 shows a schematic side view of one embodiment of an alternative unlocking mechanism,
- (18) FIG. 16A shows a schematic perspective view of one embodiment of blocking elements,
- (19) FIG. 16B shows a schematic perspective view of one embodiment of blocking elements,
- (20) FIG. 17A shows a schematic perspective view of one embodiment of blocking elements,
- (21) FIG. 17B shows a schematic perspective view of one embodiment of blocking elements,
- (22) FIG. 18A shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,
- (23) FIG. 18B shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,
- (24) FIG. 18C shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,
- (25) FIG. 18D shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,
- (26) FIG. 18E shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,
- (27) FIG. 18F shows a schematic side view of one embodiment of a portion of the seat in a movement sequence,
- (28) FIG. 19A shows a schematic side view of one embodiment of a cable mechanism in a

movement sequence,

(29) FIG. **19B** shows a schematic side view of one embodiment of the cable mechanism in a movement sequence,

(30) FIG. **19C** shows a schematic side view of one embodiment of the cable mechanism in a movement sequence,

(31) FIG. **20A** shows a schematic side view of one embodiment of a cable arresting means in a movement sequence,

(32) FIG. **20B** shows a schematic side view of one embodiment of the cable arresting means in a movement sequence,

(33) FIG. **20C** shows a schematic side view of one embodiment of the cable arresting means in a movement sequence,

(34) FIG. **20D** shows a schematic side view of one embodiment of the cable arresting means in a movement sequence,

(35) FIG. **20E** shows a schematic side view of one embodiment of the cable arresting means in a movement sequence, and

(36) FIG. **20F** shows a schematic side view of one embodiment of the cable arresting means in a movement sequence.

DETAILED DESCRIPTION

(37) Parts which correspond to one another are denoted by the same reference designations in all figures.

(38) FIG. **1** shows schematically, in a perspective illustration, a seat **1**, in particular vehicle seat, with a seat longitudinal adjustment device. The seat **1** is arranged, for example, on a vehicle body, not illustrated specifically, by the seat longitudinal adjustment device.

(39) A vehicle seat **1** which is schematically illustrated in FIG. **1** is described below using three spatial directions which run perpendicularly to one another. In the case of a seat **1** installed in the vehicle, a longitudinal direction X runs largely horizontally and preferably parallel to a vehicle longitudinal direction which corresponds to the normal direction of travel of the vehicle. A transverse direction Y which runs perpendicularly to the longitudinal direction X is likewise oriented horizontally in the vehicle and runs parallel to a vehicle transverse direction. A vertical direction Z runs perpendicularly to the longitudinal direction X and perpendicularly to the transverse direction Y. In the case of a vehicle seat **1** which is installed in the vehicle, the vertical direction Z runs preferably parallel to a vehicle vertical axis.

(40) The positional and directional indications used, such as for example front, rear, top and bottom, relate to a viewing direction of an occupant seated in the seat **1** in a normal seating position, wherein the seat **1** is installed in the vehicle, in a use position suitable for passenger transport, with an upright backrest, and is oriented in the conventional manner in the direction of travel. The seat **1** may also be installed or moved, however, in a differing orientation, for example transversely with respect to the direction of travel.

(41) The seat longitudinal adjustment device is designed as a rail arrangement **2**. The rail arrangement **2** comprises, for example, two pairs of rails **3** which are arranged at a distance from one another. The respective pair of rails **3** comprises an upper rail **31**, also called rail runner or seat rail, and a lower rail **32**, also called guide rail or floor rail. The upper rail **31** is arranged on the lower rail **32** so as to be longitudinally adjustable between a front end position and a rear end position. This adjustability permits a longitudinal adjustment of the position of the seat **1** in the longitudinal direction X, with the frontmost position of the seat **1** being assigned to the front end position of the upper rail **31** and the rearmost position of the seat **1** being assigned to the rear end position of the upper rail **31**. The upper rail **31** and the lower rail **32** each have associated rail ends **33**.

(42) The rail arrangement **2** furthermore comprises at least one locking element **4**. Each pair of rails **3** here can comprise an associated locking element **4**, for example in the form of a latching plate or

locking plate.

(43) For the releasable fastening of the seat **1** on the rail arrangement **2**, various embodiments of a modular fastening arrangement **5** are described below. The respective fastening arrangement **5** can be preassembled. In particular, the fastening arrangement **5** can be designed as a preassemblable docking mechanism **7**. Such a modular preassemblable docking mechanism **7** permits simple and rapid installation or removal of the seat **1** on or from the rail arrangement **2**.

(44) FIG. **2** shows a schematic illustration of the modular fastening arrangement **5** of the seat **1** for fastening the seat **1** on the rail arrangement **2**. The fastening arrangement **5** comprises at least one lock module **51** for releasably arresting the fastening arrangement **5** on the rail arrangement **2**, one lock unlocking module **52** for unlocking the lock module **51** and one rail unlocking module **53** for unlocking the locking element **4**.

(45) Such a modular fastening arrangement **5** designed in particular as a docking mechanism **7** can be used to simply and rapidly mount and remove the seat **1** on and from the seat longitudinal adjustment device, in particular on and from the rail arrangement **2**. The individual modules, such as the lock module **51**, the lock unlocking module **52** and the rail unlocking module **53**, can be preassemblable here. The lock module **51** in turn can comprise a plurality of separate lock units **511**. The separate lock units **511** can be coupled in terms of movement; in particular, they can be actuatable independently of one another and/or dependently on one another.

(46) Such a modular fastening arrangement **5** has few preassembled installation parts and is mountable particularly simply and rapidly. Furthermore, the lock unlocking module **52** and the rail unlocking module **53** are operable in a simple manner independently of one another or synchronously, in particular can be brought from an arresting position of a seat lock, in particular the lock units **511** of the lock module **51**, and/or of a rail locking means, in particular the locking element **4**, into a released position of the seat lock or of the rail locking means, or vice versa.

(47) The lock module **51** can be of multi-part design and comprises, for example, four lock units **511**, e.g. two pairs of lock units **511**. In one possible embodiment, the lock units **511** are operable or actuatable synchronously or independently of one another using the lock unlocking module **52**.

(48) Furthermore, the various fastening arrangements **5** can comprise an actuating module **54** for actuating the associated lock unlocking module **52** and the rail unlocking module **53**. The actuating module **54** can comprise one or more actuating units **541** for independently or dependently actuating the lock unlocking module **52** and/or the rail unlocking module **53**. The actuating units **541** can be formed separately or can be an integral part of the lock unlocking module **52** and/or of the rail unlocking module **53**. However, the actuating units **541** can also be coupled in terms of movement. The actuating units **541** can be designed as a lever arrangement, in particular as a lever arm arrangement, and/or as a tension arrangement, in particular tension strap arrangement or tension belt arrangement.

(49) FIG. **3** schematically shows, in an exploded illustration, a first embodiment of a first modular fastening arrangement **50**.

(50) The first modular fastening arrangement **50** comprises, as lock module **51**, a frame **50.1**, on which four lock units **511** are arranged and held. The frame **50.1** is, for example, a rectangular carrier or an H-shaped carrier bearing a respective lock unit **511** at its corners or at its free ends. Furthermore, the frame **50.1** serves for fastening the seat **1**. For this purpose, four fastening elements **50.3** are provided on the frame **50.1** and can be used to connect the seat **1** fixedly to the fastening arrangement **50**.

(51) The lock unlocking module **52** comprises at least one first carrier **521**, a lock actuating unit **522**, which is designed as an integral actuating unit **541**, and a lock unlocking mechanism **523**. The first carrier **521** of the lock unlocking module **52** is designed as a longitudinal rod or longitudinal tube. The lock actuating unit **522** is designed as a lever arm. The lock actuating unit **522** comprises a gripping element **522.3**. The lock unlocking mechanism **523** is designed, for example, as a pull mechanism, in particular a cable pull mechanism, which is coupled, in particular coupled in terms

of movement, to the lock units **511** in the assembled state. The lock unlocking mechanism **523** has, for example, two cable pulls **523.1** and **523.2** which are both designed as double cable pulls.

(52) The rail unlocking module **53** comprises at least one second carrier **531**, a rail actuating unit **532**, which is designed as an integral actuating unit **541**, and a rail unlocking mechanism **533**. The rail actuating unit **532** is designed as a lever arm. The rail unlocking mechanism **533** is designed as a pull mechanism or lever mechanism, in particular a cable pull mechanism or lever arm mechanism, which, in the assembled state, is coupled, in particular coupled in terms of movement, to the locking element **4** for unlocking the rail or locking the rail.

(53) Depending on the design of the rail locking means, one locking element **4** can be provided per pair of rails **3**. Alternatively, just one locking element **4** in one of the pairs of rails **3** can also be provided per rail arrangement **2**.

(54) Furthermore, for coupling the lock unlocking module **52** to the lock units **511**, the first fastening arrangement **50** can comprise one or two lock adapters **50.2** which can be arranged on the frame **50.1**.

(55) Furthermore, the respective lock adapter **50.2** can comprise guide elements or sliding elements, in particular rolling elements **6**. In the assembled state of the modular fastening arrangement **5** or **50** and in the fastening arrangements **500** and **5000** described below, the guide elements or sliding elements serve for guiding the seat **1** on or in the rail arrangement **2**, in particular on guide surfaces or rolling surfaces of the lower rails **32** of the pairs of rails **3**. The seat **1** is arranged movably in a sliding manner on the lower rail **31** in the unlocked state of the rail unlocking module **53** using the rolling elements **6** via the respective fastening arrangement **5**, **50**, **500**, or **5000**, which is mounted and arrested on the upper rails **31** by the lock module **51**.

(56) In the exemplary embodiment which is shown, the rolling elements **6** are arranged in a sliding manner on sliding surfaces or guide surfaces of the lower rails **32** (also see in the detail in FIG. **11**). Alternatively, the rolling elements **6** can also be arranged in a sliding manner within the pairs of rails **3**, in a cavity formed between upper rail **31** and lower rail **32**.

(57) FIGS. **4** and **5** schematically show, in an exploded illustration and in a perspective view, a second embodiment of a modular fastening arrangement **500** (called second fastening arrangement **500** below).

(58) Said second fastening arrangement **500** can be preassembled. The second fastening arrangement **500** can be designed as a preassembled docking mechanism **7**.

(59) The carrier or frame **50.1** and the lock adapter **50.2** are omitted in this embodiment. Said second fastening arrangement **500** therefore comprises fewer parts and modules. In these FIGS. **4** and **5**, the rail arrangement **2** is not illustrated for the sake of clarity.

(60) The second modular fastening arrangement **500** comprises, as lock module **51**, four lock units **511** with integrated carrier elements **512**. The respective lock unit **511** comprises a carrier plate **511.12**. The integrated carrier elements **512** serve for fastening the seat **1** in a manner not illustrated specifically. The carrier elements **512** are designed, for example, as fastening bolts or fastening pins which are connected to the seat **1** and to the carrier plate **511.12** in the assembled state.

(61) In the assembled state with the rail arrangement **2**, the second fastening arrangement **500** is fixed in a form-fitting and/or force-fitting manner at the rail ends **33** of the upper rails **31** by a respective lock unit **511**.

(62) The four lock units **511** each comprise a blocking element **513** for releasable locking in the rail arrangement **2**. The respective blocking element **513** is designed, for example, as a blocking hook. The blocking hooks are used to releasably fix and fasten the second fastening arrangement **500** on the rail arrangement **2**.

(63) Furthermore, each lock unit **511** comprises an unlocking unit **514** for coupling the lock unit **511** to the actuating module **54**, in particular to one of the actuating units **541**. The unlocking unit **514** is designed as a joint mechanism and/or lever mechanism.

(64) The respective unlocking unit **514** can be used to unlock the associated lock unit **511**, in

particular to unlock all the lock units **511** synchronously, such that the lock module **51** can be opened and the second fastening arrangement **500** extracted or removed from the seat **1**.

(65) The unlocking unit **514** is coupled in terms of movement to the rail unlocking module **53** such that, when the rail unlocking module **53** is actuated in order to unlock the locking element **4** of the rail arrangement **2**, the unlocking units **514** are actuated synchronously in order to unlock the lock units **511** of the lock module **51**. As a result, by actuation of the rail unlocking module **51**, for example by an associated rail actuating unit **532**, the locking elements **4** of the rail arrangement **2** and the lock units **511** of the lock module **51** can be unlocked synchronously, and therefore the fastening arrangement **500**, and thus optionally the seat **1**, is released from the rail arrangement **2** and can be extracted.

(66) Two lock units **511**, in particular a front lock unit **511** and a rear lock unit **511**, as seen in the longitudinal direction X of the rail arrangement **2**, are provided per pair of rails **3**. The blocking elements **513** are mounted pivotably on the lock units **511**. The blocking elements **513** are designed as blocking hooks **517**.

(67) The two blocking elements **513** associated with a pair of rails **3** are mounted on the associated lock units **511** in such a manner that the blocking hooks **517** thereof are directed towards one another. In particular, when the unlocking unit **514** is actuated in order to unlock and open the lock units **511**, the two blocking elements **513** of the respective pair of rails **3** are pivoted away from one another. The blocking hooks **517** thereby pivot out of their arresting engagement at the rail ends **33**. The lock module **51** is therefore opened and the second fastening arrangement **500** can be released and extracted from the form-fitting and/or force-fitting connection to the rail arrangement **2**.

(68) The lock unlocking module **52** comprises at least one first carrier **521**, a lock actuating unit **522**, which is designed as an integral actuating unit **541**, and a lock unlocking mechanism **523**. The first carrier **521** of the lock unlocking module **52** is designed as a transverse rod or a transverse tube.

(69) The lock actuating unit **522** is designed as a lever arm mechanism. The lever arm mechanism comprises two lever arms **522.1** and **522.2**, wherein one of the lever arms **522.1** is provided with a gripping element **522.3**. The two lever arms **522.1** and **522.2** are arranged on the carrier **521** for rotation therewith and are coupled in terms of movement via the first carrier **521** such that, when the lever arm **522.1** is actuated, the grippless lever arm **522.2** is entrained by the gripping element **522.3** as a result of the rotation of the first carrier **521**. The lever arms **522.1** and **522.2** are arranged perpendicularly to the first carrier **521**.

(70) The lock unlocking mechanism **523** is designed as a pull mechanism, in particular a cable pull mechanism, which is coupled, in particular coupled in terms of movement, to the lock units **511** in the assembled state. The pull mechanism has two cable pulls **523.1** and **523.2** which are both designed as double cable pulls. The two cable pulls **523.1** and **523.2** are respectively coupled at one end to one of the lever arms **522.1** and **522.2** and are coupled at the other end to two associated lock units **511**. By actuation of the lever arm **522.1** and by the other lever arm **522.2** being entrained, the two cable pulls **523.1** and **523.2** are actuated, thus unlocking the associated lock units **511**, in particular the blocking elements **513**. The lock unlocking mechanism **523** is designed as a separate mechanism for unlocking the lock module **51**, in particular for synchronously unlocking all of the lock units **511**.

(71) The rail unlocking module **53** comprises at least one second carrier **531**, a rail actuating unit **532**, which is designed as an integral actuating unit **541**, and a rail unlocking mechanism **533**.

(72) The rail actuating unit **532** is designed as a lever arm. The rail unlocking mechanism **533** is designed as a pull mechanism, articulated mechanism and/or bar mechanism, which, in the assembled state, is coupled, in particular coupled in terms of movement, to the locking element **4** for unlocking the rail or locking the rail, and/or to the lock units **511**. In order to actuate the rail unlocking mechanism **533**, the associated integral actuating unit **541** comprises a gripping unit **534**.

(73) Depending on the design of the rail locking means, one locking element **4** can be provided per

pair of rails **3**. Alternatively, just one locking element **4** in one of the pairs of rails **3** can also be provided per rail arrangement **2**. Accordingly, the rail unlocking mechanism **533** can comprise one or two rail unlocking units **533.1**.

(74) In order to actuate the rail unlocking mechanism **533**, the rail unlocking module **53** comprises one rail unlocking unit **533.1** per pair of rails **3**. The respective rail unlocking unit **533.1** comprises two lever arms **531.1** which emerge perpendicularly from the carrier **531** and on each of which a free end of a rail unlocking arm **533.2** is formed which actuates the locking element **4** of the respective pair of rails **3** for unlocking or locking purposes.

(75) During the removal or final installation and fastening of the seat **1** from or on the rail arrangement **2**, in said second fastening arrangement **500** the lock unlocking module **52** and the rail unlocking module **53** are coupled to each other in terms of movement. For this purpose, two of the unlocking units **514** are each coupled in terms of movement to one of the rail unlocking units **533.1** of the rail unlocking mechanism **533**.

(76) The unlocking units **514** are designed, for example, as lever arms **514.1** or articulated arms. The rail unlocking units **533.1** are designed as vertically movable pins **533.3** or bolts. Two lever arms **514.1** of the two unlocking units **514** for unlocking the locking element **4** of one of the pairs of rails **3** are coupled to, in particular held pivotably on, the pin **533.3** of one of the rail unlocking units **533.1**.

(77) Such an arrangement permits the synchronous actuation of the rail unlocking module **53**, of the lock module **51**, in particular all of the lock units **511**, and of the lock unlocking module **52**, in particular all of the unlocking units **514**, such that the fastening arrangement **500** can be removed from or mounted on the rail arrangement **2**.

(78) In order to adjust the blocking element **513** from the locking position into the unlocking position, the respective lock unit **511** comprises a control element **511.2**. The control element **511.2** has a guide slot **511.21** (illustrated in FIGS. **5**, **6** and **8**) into which a guide pin **511.8** (illustrated in FIGS. **5** and **6**), which is arranged on the blocking element **513**, is guided.

(79) The construction and the components of the respective lock unit **511** are described in more detail below with reference to FIG. **8**.

(80) FIGS. **6** and **7** schematically show, in an exploded illustration and in a perspective view, a third embodiment of a modular fastening arrangement **5000** (called third fastening arrangement **5000** below). The carrier or frame **50.1** and the lock adapter **50.2** of the first fastening arrangement **50** are omitted in this embodiment. Said third fastening arrangement **5000** therefore comprises fewer parts and modules. In these FIGS. **6** and **7**, the rail arrangement **2** is not illustrated for the sake of clarity.

(81) The third modular fastening arrangement **5000** comprises, as lock module **51**, four lock units **511** with integrated carrier elements **512**. The integrated carrier elements **512** serve for fastening the seat **1** in a manner not illustrated specifically. The third modular fastening arrangement **5000** differs in the type of lock unlocking means which comprises a cable pull mechanism instead of an articulated mechanism as the unlocking unit **514**.

(82) The four lock units **511** each comprise, analogously to the second modular fastening arrangement **500**, a blocking element **513** for the releasable locking on or in the rail arrangement **2**. The respective blocking element **513** is designed, for example, as a blocking hook **517**.

(83) Furthermore, a pair of lock units **511** is in each case coupled in terms of movement to one another by an associated unlocking unit **514**, for example unlocking lever or unlocking arm. For example, the unlocking unit **514** is designed as a deflecting pulley **515** which is actuable by the lock actuating unit **522**. Instead of a lever arm **522.1**, **522.2**, the lock actuating unit **522** comprises a lock actuating cable **522.4**. A gripping element **522.3**, in particular a cable loop, is arranged at the free end of the lock actuating cable **522.4**.

(84) The lock unlocking module **52** comprises, as first carrier **521**, a transmission rod, the lock actuating unit **522**, which is designed as an integral actuating unit **541**, and the lock unlocking

mechanism **523**. The first carrier **521** of the lock unlocking module **52** is designed as a transverse rod or a transverse tube.

(85) The lock actuating unit **522** is designed as a cable pull mechanism. The cable pull mechanism comprises the lock actuating cable **522.4** which is provided with the gripping element **522.3**. The lock actuating cable **522.4** is arranged via the deflecting pulley **515** on the first carrier **521** for rotation therewith, such that, when the lock actuating cable **522.4** is actuated, the first carrier **521** is rotated at the same time by the gripping element **522.3** in order to actuate the unlocking unit **514**, in particular the deflecting pulley **515**, which, in turn, is coupled, in particular coupled in terms of movement, to the lock unlocking mechanism **523**.

(86) The lock unlocking mechanism **523** is designed as a pull mechanism, in particular a cable pull mechanism, which is coupled, in particular coupled in terms of movement, to the lock units **511** in the assembled state. The cable pull mechanism has two cable pulls **523.1** and **523.2** which are both designed as a lock unlocking cable. The two cable pulls **523.1** and **523.2** are respectively coupled at one end to one of the front lock units **511** and are coupled at the other end to one of the rear lock units **511**. For this purpose, each lock unit **511** has an associated winding unit **524**.

(87) By actuation of the lock actuating cable **522.4** and driving the deflecting pulley **515**, the two cable pulls **523.1** and **523.2** are actuated, as a result of which the associated lock units **511** are unlocked or locked by winding said cable pulls onto or unwinding them from the winding units **524**.

(88) The rail unlocking module **53** comprises at least the second carrier **531**, the rail actuating unit **532**, which is designed as an integral actuating unit **541**, and the rail unlocking mechanism **533** (not illustrated specifically in FIGS. **6** and **7**, and constructed analogously as shown in FIGS. **4** and **5** and described with regard thereto).

(89) The rail actuating unit **532** is designed as a lever arm. The rail unlocking mechanism **533** is designed as a pull mechanism, articulated mechanism and/or bar mechanism, which, in the assembled state, is coupled, in particular coupled in terms of movement, to the locking element **4** for unlocking the rail or locking the rail, and/or to the lock units **511**. In order to actuate the rail unlocking mechanism **533**, the associated integral actuating unit **541** comprises the gripping unit **534**.

(90) Depending on the design of the rail locking means, one locking element **4** can be provided per pair of rails **3**. Alternatively, just one locking element **4** in one of the pairs of rails **3** can also be provided per rail arrangement **2**. Accordingly, the rail unlocking mechanism **533** can comprise one or two rail unlocking units **533.1**.

(91) During the removal or final installation and fastening of the seat **1** from or on the rail arrangement **2**, in said third fastening arrangement **5000** the lock unlocking module **52** and the rail unlocking module **53** are coupled in terms of movement to one another. For this purpose, the respective deflecting pulley **515** is coupled in terms of movement to the rail unlocking mechanism **533**, in particular is coupled in terms of movement to an associated rail unlocking arm **533.2**.

(92) FIG. **8** schematically shows, in an exploded illustration, an exemplary embodiment of a lock unit **511**. All of the lock units **511** of the lock module **51** are constructed analogously.

(93) The respective lock unit **511** comprises at least: one first spring element **511.1** (also called “spring control element”) for a control element **511.2** (also called “control element”) one second spring element **511.3** (also called “spring crash cam”), two carrier elements **512** (also called “bolt cam”) one compensating spring **511.4** (also called “spring tolerance cam”) for compensating for tolerances between the carrier elements **512**, one blocking hook holder **511.5** (also called “bolt hook”) for the blocking hook **517**, one holding pin **511.6** (also called “pin control element”) for the control element **511.2**, the unlocking unit **514** (also called “link track rod”), one coupling pin **511.7** (also called “pin cam link”), the blocking element **513** with the blocking hook **217** (also called “crash hook”), two guide pins **511.8** (also called “pin ISO 8734 6×20”), one compensating element **511.9** (also called “tolerance cam”), one crash cam element **511.10** (also called “crash cam”), one

reinforcing element **511.11** (also called “reinforcement plate”), optionally for reinforcing the carrier plate **511.12**, one carrier plate **511.12** (also called “side plate”), one triggering plate **511.13** (also called “release plate”), one third spring element **511.14** for a triggering arm (also called “spring release lever”), and one bearing bushing **511.15** for the third spring element **511.14** (also called “bush spring”).

(94) FIG. **9** shows the lock unit **511** preassembled in the assembled state as a preassembly unit.

(95) The triggering plate **511.13** comprises an entraining pin **511.16** as a guide pin **511.8**.

Furthermore, the triggering plate **511.13** comprises a receptacle **511.17**, in particular a slot, for receiving and holding one end of the associated cable pull **523.1**. The third spring element **511.14** is designed as a restoring spring.

(96) The respective lock module **51** can be actuated differently: actuation by the lock unlocking mechanism **523** or actuation via the unlocking unit **514** by the rail unlocking module **53** and thus synchronous actuation of the rail unlocking mechanism **533** for the locking elements **4** of the rail arrangement **2** and of the unlocking units **514** for the lock units **511**.

(97) When the cable pull **523.1** is actuated to unlock the lock unit **511**, in particular to pivot the blocking hook **517** and therefore to release the blocking hook **517** from the upper rail **31**, the triggering plate **511.13** is entrained and moved, in particular rotated, by the entraining pin **511.16**. As a result, the third spring element **511.14** is tensioned and the blocking element **513** and therefore the blocking hook **517** are pivoted. The blocking hook **517** is positioned into an unlocking position and thus released from the upper rail **31**.

(98) If the actuation is ended, the triggering plate **511.13**, the blocking element **513** and the blocking hook **517** are reset by the third spring element **511.14** into their starting position which corresponds to a locking position.

(99) During the actuation of the unlocking unit **514** by the rail unlocking module **53**, in particular the rail unlocking unit **533.1**, the crash cam element **511.10** is forcibly guided, in particular rotated by the guide pin **511.8** about an axis of rotation **511.18**. The guide pin **511.8** arranged on the cam element **511.10** is guided in a guide slot **511.19**. For this purpose, the crash cam element **511.10** is connected in an articulated manner to the unlocking unit **514** at the point of articulation **511.20**. The crash cam element **511.10** entrains the blocking element **513** by the coupling via the compensating element **511.9** and the control element **511.2**. The blocking element **513** is moved from the locking position into the unlocking position. For this purpose, the control element **511.2** has the guide slot **511.21** (illustrated in FIGS. **5**, **6** and **8**) into which the guide pin **511.8** (illustrated in FIGS. **5** and **6**), which is arranged on the blocking element **513**, is guided when the blocking element **513** is moved from the locking position into the unlocking position.

(100) FIG. **10** shows a schematic illustration of a releasable seat **1** on the rail arrangement **2**. In order to install the seat **1**, the latter is to be arranged on the rail arrangement **2**, in a region in which the lower rail **32** is free of the upper rail **31**.

(101) FIG. **11** shows a schematic illustration of the seat **1** pre-positioned on the rail arrangement **2**. The seat **1** is pre-positioned here by rolling elements **6** on its lower side on the lower rail **32**. The lock module **51** is open. The seat **1** is guided on sliding surfaces of the lower rail **32** (or alternatively of the upper rail **31**, not illustrated) by the rolling elements **6** and pre-positioned in such a manner that the lock units **511** are positioned above the rail ends **33** of the upper rail **31**.

(102) FIGS. **12** to **20F** show schematic illustrations of various embodiments and the movement sequences thereof during the unlocking or locking of one of the fastening arrangements **5**, **50**, **500**, **5000** from or on the rail arrangement **2**.

(103) FIG. **12** shows, as actuating module **54**, a lever arrangement which, in order to release the fastening arrangement **5**, **50** or **500**, is actuable downwards in the direction of the rail arrangement **2**, according to arrow **P1**, in order to allow the blocking elements **513** of the lock units **511** to drop into the lower rails **32** of the rail arrangement **2**, in particular into a receiving slot of the U-shaped lower rails **32**. For the description of the fastening arrangements **5**, **50** and **500**, reference is made to

FIGS. 1 to 5 and 8 to 11.

(104) In FIG. 12, the fastening arrangement 5, 50 or 500 is shown in the unlocked or open state. The lever arms 514.1 and therefore the blocking elements 513 are raised here as a result of the upwardly actuated actuating module 54, in particular the actuating unit 541. In this state, the seat 1 with the fastening arrangement 5, 50 or 500 can be pre-positioned on the lower rail 32 by the rolling elements 6.

(105) The seat 1 comprises two lateral seat carriers 1.1. Each pair of rails 3 is assigned a seat carrier 1.1.

(106) The lock units 511 with the blocking elements 513 are each arranged here on an outer side of the seat carriers 1.1. The rolling elements 6 are arranged on an inner side of the seat carriers 1.1. The actuating module 54 is arranged between the two seat carriers 1.1 and connected via passage openings in the seat carriers 1.1 to the outer lock module 51. Components of the lock unlocking module 52 and/or of the rail unlocking module 53 are partially or completely arranged on the inner side and/or the outer side of the seat carriers 1.1.

(107) When the actuating unit 541 is actuated downwards according to arrow P1, the blocking elements 513, in particular the blocking hooks 517, then engage in an arresting manner on the rail ends 33 of the upper rail 31. In particular, the blocking elements 513 of the lock units 511 drop into the respective receiving slot of the U-shaped lower rails 32 and reach under the rail ends 33 of the upper rail 31 such that the seat 1 is fastened in a vertically secured manner to the rail arrangement 2.

(108) FIG. 13 shows, as actuating module 54, the previously described cable pull mechanism which, in order to release the third fastening arrangement 5000, is actuable upwards away from the rail arrangement 2, according to arrow P2, in order to release the blocking elements 513 of the lock units 511 from the rail arrangement 2.

(109) In order to arrest the blocking hooks 517 on the respective upper rail 31, the respective upper rail 31 comprises a hook receptacle 31.1 at each of its rail ends 33. The upper rail 31 is arranged movably in the lower rail 32 by further rolling elements 6, in particular in each case two rolling elements 6 in the region of each rail end 33. In particular, the hook receptacles 31.1 form the respective rail end 33. The rolling elements 6 are arranged in the upper rail 31 between the two hook receptacles 31.1, in particular substantially adjacent to them, and roll on a base surface of the lower rail 32.

(110) For the description of the fastening arrangements 5 and 5000, reference is made to FIGS. 1, 6 and 7, and 10 to 11.

(111) FIG. 14A shows a side view of the fastening arrangement 5, 50 or 500 with the docking mechanism 7 in order to mount or to remove the respective fastening arrangement 5, 50 or 500 on or from the rail arrangement 2. The side view means that only half of the fastening arrangement 5, 50 or 500 is shown with the two lock units 511 and the lock unlocking mechanism 523 with one of the cable pulls 523.1 for one of the pairs of rails 3. The other part of the docking mechanism 7 for the other pair of rails 3 is constructed analogously, as previously described. The actuating module 54 is configured in such a manner that it can actuate the two parts synchronously, as has been previously described in more detail with reference to FIGS. 1 to 5 and 8 to 13.

(112) FIG. 14A shows, in the side view, that side of the carrier plate 511.12 of the respective lock unit 511 which forms an outer side in the mounted state on the seat carriers 1.1.

(113) FIG. 14B shows an unlocking mechanism 8 which is provided on the opposite side of the carrier plate 511.12 of the respective lock unit 511 in order to arrest the respective fastening arrangement 5, 50 or 500 on the rail arrangement 2 for the fastening of the seat 1 or to release same therefrom and to unlock the rail locking means, in particular the locking elements 4 of the pairs of rails 3, synchronously.

(114) The unlocking mechanism 8 is designed as a lever mechanism and/or articulated mechanism for unlocking the rails in combination with a cable mechanism for unlocking the lock.

(115) FIG. 15 shows an alternative unlocking mechanism 80 which is designed as a cable pull mechanism and corresponds to the previously described third fastening arrangement 5000. FIG. 15 shows the fastening arrangement 5000 which is arranged between the seat 1 and the rail arrangement 2 and releasably locks the seat 1 to the rail arrangement 2 using the lock units 511.

(116) FIGS. 16A and 16B show the blocking elements 513 of two lock units 511 on one of the pairs of rails 3 being unlocked by a lever mechanism corresponding to the unlocking mechanism 8.

(117) By pulling the rail unlocking arm 533.2 upwards in accordance with arrow P3, the rail unlocking mechanism 533 is actuated, in particular the second carrier 531, which is designed as an unlocking rod, is rotated such that the unlocking units 514 are actuated in accordance with arrow P4 and the locking element 4 is unlocked, and thus the rails 31, 32 are released. The release of the rails 31, 32 in the “vis-à-vis” position leads to increased complexity since the unlocking rod has to be rotated in the same direction both on the inner side and on the outer side of the carrier plate 511.12, with the pulling direction on one side being switched over in the opposite direction.

(118) FIGS. 17A and 17B show the blocking elements 513 of two lock units 511 on one of the pairs of rails 3 being unlocked by a cable pull mechanism corresponding to the unlocking mechanism 80 for the third fastening arrangement 5000. The rail actuating unit 532 is pivoted or pulled upwards in accordance with arrow P5. As a result, the unlocking unit 514 is moved, in particular pivoted, downwards, in accordance with arrow P6, in order to unlock the rail locking means, in particular the locking element 4.

(119) The direct unlocking of the rail arrangement 2 is undertaken by pressing down the locking element 4, for example an unlocking plate, vertically in the upper rail 31 (also called rail runner) and reduces the complexity and possible problems when latching the unlocking unit 514, for example an unlocking lever, unlocking arm or unlocking pin, into an unlocking bushing of the upper rail 31 (rail runner) during the docking operation. The triggering mechanism or unlocking mechanism according to circle K functions both in the “forwardly directed” position and in the “vis-à-vis” position.

(120) FIG. 18A shows the seat 1 which rolls along the lower rail 32 in the direction of the upper rail 31 in accordance with arrow P7. The blocking elements 513 are guided in slot-shaped openings in the U-shaped lower rail 32.

(121) FIG. 18B shows the lifting of the front blocking element 513 of the front lock unit 511 out of the lower rail 32, wherein, during the further movement of the seat 1 in the direction of the preassembled upper rail 31, the blocking element 513 is pushed onto the upper rail 31 and moved along the latter.

(122) FIG. 18C shows the state of the fastening arrangement 5 (50, 500 or 5000) after the seat 1 has been moved forward to such an extent that the upper rail 31 strikes against the rear blocking element 513, wherein, owing to restoring elements, the front blocking element 513 latches again automatically into the upper rail 31 in accordance with arrow P8 and therefore the two lock units 511 are fully latched into place. In this position, the rail unlocking unit 533.1, in particular a track rod, drops into the upper rail 32 (also called running rail or track slider) such that an arresting connection in accordance with arrow P9 is produced and the seat 1 is arrested on the rail arrangement 2 by the fastening arrangement 5, 50 or 500 and the locking element 4 that has been latched into place.

(123) FIGS. 18D and 18E show an actuation of the lock actuating unit 522 in the upwards direction in accordance with arrow P10 firstly for unlocking the rails via the actuated unlocking units 514, in particular unlocking levers. The unlocking units 514, in particular unlocking levers, are connected via cable pulls 523.1, in particular via Bowden cables, to the locking element/locking elements 4, in particular locking plates, and lift them in accordance with arrows P11 and P12. In the process, the triggering plates 511.13 of the lock units 511 are rotated and lift the control elements 511.2 and, with the latter, the blocking elements 513, upwards until the blocking elements 513 are released from the locking region on the upper rail 31 and are therefore unlocked. In parallel therewith, the

locking of the rails is released by the locking element **4** being moved, in particular lifted, into a release position and out of the upper rail **31** and the lower rail **32**.

(124) FIG. **18F** shows the further actuation of the lock actuating unit **522** into a further raised position, wherein the triggering plates **511.13** are rotated further and the control elements **511.2** and the blocking elements **513** are raised further until the blocking elements **513** are arranged above the respective pair of rails **3** in accordance with arrow **P13**. In this release position or unlocking position of the fastening arrangement **5**, **50**, **500** or **5000**, the lock actuating unit **522** is held secured in the associated raised position by the third spring element **511.14** (also called softlock spring). The seat **1** can now be rolled on the rail arrangement **2** or removed from the latter.

(125) FIGS. **19A** to **19C** show a cable mechanism as actuating module **54**.

(126) FIG. **19A** shows an actuation of the lock actuating cable **522.4** upwards in accordance with arrow **P14**. The deflecting pulley **515** (also called central pulley) is thereby moved in the anticlockwise direction, in particular rotated in accordance with arrow **P15**.

(127) FIG. **19B** shows that, as a result of the rotational movement of the deflecting pulley **515**, the cable pulls **523.1** and **523.2** are actuated and raise a cam disc **516**, in particular pivot the latter upwards, in accordance with arrow **P16**.

(128) FIG. **19C** shows that, upon further rotation of the deflecting pulley **515**, the blocking elements **513** and the control elements **511.2** are raised in accordance with arrow **P17**. In order to hold the fastening arrangement **5** or **5000** in this release position or unlocked position, said fastening arrangement comprises a cable arresting means **9** (also called pull-pull lock) which arrests the cable pulls **523.1** and the lock actuating cable **522.4**. The cable arresting means **9** is designed as a rotatable arresting hook or blocking lug **9.1**.

(129) FIGS. **20A** to **20F** show the sequence of movement of the cable arresting means **9**.

(130) FIG. **20A** shows the deflecting pulley **515** which rotates during the unlocking operation in accordance with arrow **P18** and, in the process, entrains the cable arresting means **9**, in accordance with arrow **P19**, by an entraining pin **10**, which also serves as arresting pin **9.2**.

(131) FIG. **20B** shows the cable arresting means **9** in the arresting position in which it arrests the cable pulls **523.1** and the lock actuating cable **522.4** and the fastening arrangement **5**, **5000** and therefore the blocking elements **513** are placed in the fully released position (release position). The cable arresting means **9** springs back until its blocking lug **9.1** strikes against an arresting pin **9.2**, in accordance with arrow **P20**.

(132) FIG. **20C** shows the cable arresting means **9** in a first secured position **GS1** when the lock actuating cable **522.4** is released again. The seat **1** can be moved, in particular rolled along the upper rail **31**, or can be removed from the rail arrangement **2**.

(133) FIG. **20D** shows the cable arresting means **9** in a second secured position **GS2** in accordance with arrow **P21**, into which it is brought when the lock actuating cable **522.4** is actuated a further time. For this purpose, the blocking lug **9.1** has a blocking contour **9.3** which has correspondingly shaped blocking receptacles for the first and second secured position **GS1**, **GS2** for the arresting pin **9.2**.

(134) FIG. **20E** shows the cable arresting means **9** when the lock actuating cable **522.4** is released. In the process, the arresting pin **9.2** moves in the Y axis direction and below the cable arresting means **9** along the surface side thereof, in accordance with arrow **P22**.

(135) FIG. **20F** shows the automatic, in particular spring-supported, restoring of the lock actuating cable **522.4** and of the deflecting pulley **515** into their starting position or locking position, in which the blocking elements **513** move downwards and therefore drop into the rail arrangement **2** when the seat **1** is arranged together with the associated fastening arrangement **5**, **50**, **500** or **5000**. The cable arresting means **9** is released in the process. The arresting pin **9.2** is disengaged.

LIST OF REFERENCE DESIGNATIONS

(136) **1**. Seat **1.1** Seat carrier **2** Rail arrangement **3** Pair of rails **31** Upper rail **31.1** Hook receptacle **32** Lower rail **33** Rail end **4** Locking element **5** Fastening arrangement **51** Lock module **511** Lock

unit **511.1** First spring element **511.2** Control element **511.3** Second spring element **511.4** Compensating spring **511.5** Blocking hook holder **511.6** Holding pin **511.7** Coupling pin **511.8** Guide pin **511.9** Compensating element **511.10** Crash cam element **511.11** Reinforcing element **511.12** Carrier plate **511.13** Triggering plate **511.14** Third spring element **511.15** Bearing bushing **511.16** Entraining pin **511.17** Receptacle **511.18** Axis of rotation **511.19** Guide slot **511.20** Point of articulation **511.21** Guide slot **512** Integrated carrier element **513** Blocking element **514** Unlocking unit **514.1** Lever arm **515** Deflecting pulley **516** Cam disc **517** Blocking hook **52** Lock unlocking module **521** First carrier **522** Lock actuating unit **522.1**, **522.2** Lever arm **522.3** Gripping element **522.4** Lock actuating cable **523** Lock unlocking mechanism **523.1**, **523.2** Cable pull **524** Winding unit **523** Rail unlocking module **531** Second carrier **531.1** Lever arm **532** Rail actuating unit **533** Rail unlocking mechanism **533.1** Rail unlocking unit **533.2** Rail unlocking arm **533.3** Pin **534** Gripping unit **54** Actuating module **541** Actuating unit **50** First modular fastening arrangement **50.1** Frame **50.2** Lock adapter **50.3** Fastening element **500** Second modular fastening arrangement **5000** Third modular fastening arrangement **6** Rolling element **7** Docking mechanism **8**, **80** Unlocking mechanism **9** Cable arresting means **9.1** Blocking lug **9.2** Arresting pin **9.3** Blocking contour **10** Entraining pin **GS1** First secured position **GS2** Second secured position **K** Circle **P1** to **P22** Arrow **X** Longitudinal direction **Y** Transverse direction **Z** Vertical direction

Claims

1. A fastening arrangement of a seat on a rail arrangement, wherein the fastening arrangement is of modular construction and comprises: one lock module for releasably arresting the fastening arrangement on the rail arrangement, wherein the lock module is of multi-part design and comprises at least two lock units, wherein each of the lock units are releasably fixed to both the seat and the rail arrangement, and one unlocking module for unlocking the lock module, wherein at least two rolling elements are rotatably mounted to the lock module so that the fastening arrangement is rolled away on the rail arrangement when the lock module is unlocked.
 2. The fastening arrangement according to claim 1, wherein the lock units are arranged on a frame or on a carrier plate.
 3. The fastening arrangement according to claim 1, wherein the unlocking module is configured to unlock the lock units synchronously from the rail arrangement.
 4. The fastening arrangement according to claim 1, wherein an actuating module actuates the unlocking module of the fastening arrangement and/or a rail unlocking module.
 5. The fastening arrangement according to claim 4, wherein the actuating module is coupled to the lock module via the unlocking module.
 6. The fastening arrangement according to claim 4, wherein the actuating module is coupled to a locking element for the rail arrangement via the unlocking module.
 7. A seat with a rail arrangement and a fastening arrangement according to claim 1 for releasably fastening the seat on the rail arrangement.
 8. The seat according to claim 7, wherein the fastening arrangement is designed as a preassemble docking mechanism.
 9. The seat according to claim 7 wherein the rail arrangement comprises two pairs of rails each having an upper rail and a lower rail, and the fastening arrangement is releasably arrestable at one rail end of the upper rail by a respective lock unit.
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